NVIDIA × RED HAT SMALL MODELS HACK · MULTI-AGENT SYSTEMS TRACK

RiskLens AI

Explain every dollar of credit loss.

A team of small NVIDIA Nemotron agents that investigates why the CECL / IFRS 9 allowance moved, using only deterministic risk tools, and reconciles every explanation back to the booked number.

Team Residual Zero
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THE PROBLEM

The allowance moved $19.24M. Now explain it before the close

CECL ALLOWANCE (SYNTHETIC BANK, 50,000 LOANS)

Jul 31 **$99.06M**

Aug 31 **$118.30M**

+$19.24M

+19.4% in one month, on a book that grew 0.3%

The questions the CFO, auditors and examiners ask

1. Which portfolio caused it?
2. Which borrowers should we look at?
3. Macro forecast, credit quality or growth?
4. Did we change a model or the weights?
5. Does the explanation tie to the booked number?

Today the answer is stitched together by hand across Finance, Credit Risk, Model Risk, economists and portfolio managers.

Figures from the RiskLens synthetic bank (seed 20260831), produced by the deterministic CECL engine.

WHY EXPLAINING ECL IS HARD (CECL OR IFRS 9)

Lifetime, forward-looking losses have many moving parts at once

Lifetime and forward-looking

CECL books lifetime loss on day one; IFRS 9 adds stage moves between 12-month and lifetime ECL.

Scenarios and weights

Baseline, adverse and severe paths, re-forecast every vintage and re-weighted by committee.

Interactions

Macro, credit migration and exposure compound. A sequential walk depends on its order.

Judgment layers

Management overlays and weight changes sit beside model output and must be shown separately.

Many owners

Exposure, ratings, forecasts, models and the GL each live with a different team.

It must tie out

Auditors expect drivers to sum to the booked change. Plugged residuals hide real gaps.

Same investigation under ASC 326 (CECL) and IFRS 9. The RiskLens engine runs both frameworks on the same book.

THE MARKET

Banks already pay for calculation. Explanation is still manual

$47.3B risk analytics market in 2026, heading to $76.5B by 2031 (10.1% CAGR)[1]

70% of community banks use third-party vendors to run or support their CECL estimate[2]

57% of community banks rely on third-party economic forecasts for CECL[2]

77% of banks needed extra working days after year-end to produce IFRS 9 impairment figures[3]

Under scrutiny, explanation matters more. The Fed's Vice Chair for Supervision called for CECL repeal at a FASB roundtable in May 2026[4]; IFRS 9 stays in force abroad. Whatever the standard, the allowance and its movement still have to be explained.

1 Mordor Intelligence via ResearchAndMarkets, Risk Analytics Market (2026). 2 ABA Banking Journal, “CECL’s true costs come into focus”, Apr 27, 2026 (ABA community bank survey). 3 Deloitte, Sixth Global IFRS Banking Survey, 2016 (91 banks). 4 ABA Banking Journal, May 12, 2026.

OUR INSIGHT

Small models don't need to know everything. They need to know which risk tool to call next.

LLMs reason

Read the question, plan the investigation, delegate to specialists, pick tools, write the explanation, separate fact from interpretation.

Python computes

33 deterministic tools over the booked CECL run: attribution, migration, macro Shapley, model inventory, reconciliation.

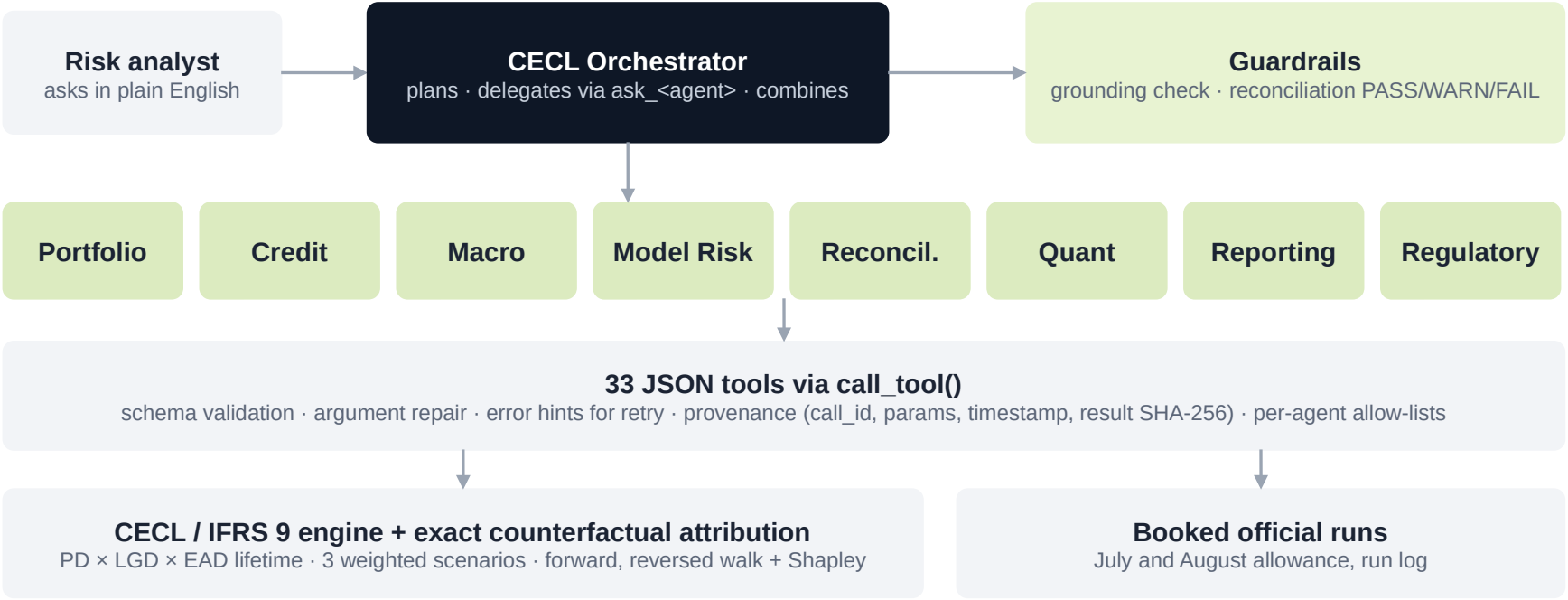
Guardrails verify

Every figure in an answer is checked against tool outputs. Every tool call is logged with a result hash.

Rule one for every agent: never compute a number yourself.

ARCHITECTURE

How RiskLens is wired



Models Nemotron 3.5 Lightning (30B total, ~3B active) on Red Hat vLLM, NVIDIA Brev A100 · Nemotron 3 Nano 4B local (Ollama, laptop) ·
Eval NeMo Gym environment + verifier

WHAT WE BROUGHT

The agents sit on a real month-end CECL pipeline, not a toy

1 Source extracts

50k loans, counterparties, macro vintages, weights, overlays

2 Pre-run controls

17 checks: data quality, model approval, weights

3 Official run

Lifetime PD × LGD × EAD, 3 scenarios weighted, CECL and IFRS 9

4 Booked allowance

Results + run log: what the tools read and reconcile to

Lessons carried over from our Dataiku ECL workbench

Plugged → exact
The old waterfall rescaled drivers to hit the total, so its residual was zero by construction. We rebuilt it as a counterfactual walk with a genuine residual, checked by a reversed walk and Shapley.

Known answer
Injected story (macro, ~20 CRE downgrades, $40M new CRE, runoff, no model change) with hidden ground truth, so agents are scored, not admired.

Real data ready
Point RISKLENS_DATA_DIR at extracts with the same columns (e.g. from Dataiku) and rerun. No code changes.

INSIDE THE DETERMINISTIC ENGINE

Atomic PD, LGD and EAD models feed one engine that books CECL and IFRS 9

INPUTS

Loan snapshot
balance, term, undrawn, rating, DSCR, LTV, FICO, DTI, days past due

Macro paths
6 variables × 3 scenarios, 8 quarters + reversion

Committee choices
scenario weights, overlays

Model inventory
which version each product uses

ATOMIC MODELS

PD Logistic scorecards + macro logit shift
CRE_PD_v4 · CI_PD_v3 · MORT_PD_v3 · AUTO_PD_v2 · CONS_PD_v2 · CC_PD_v5

LGD Collateral (LTV) + macro sensitivity, floor and cap
CRE_LGD_v3 · CI_LGD_v2 · MORT_LGD_v2 · AUTO · CONS · CC · LEASE

EAD Balance + CCF × undrawn, amortizing profile
EAD_BAL_v1 · EAD_CCF_v2 (revolvers)

CECL ENGINE

Per loan, per year, per scenario
marginal PD × LGD × EAD × discount

Sum over the loan's life
contractual term; revolvers 3 yrs; cap 30

Weight the 3 scenarios
baseline / adverse / severe

Add overlays
Checked vs a hand calc and an independent loop implementation

BOOKED RESULT

CECL (ASC 326)
Lifetime ECL for every loan, undiscounted

IFRS 9
Stage 1: 12-month · Stage 2/3: lifetime, EIR-discounted; SICR on PD, notches, DPD, watchlist

$$\text{Allowance} = \Sigma_{\text{loans}}\ \Sigma_{\text{scenarios}}\ w \cdot \Sigma_{\text{years}}\ PD \times LGD \times EAD \times DF + \text{overlays}$$

Every model ID, parameter and weight lives in config (models.yaml, methodology.yaml) and the model inventory, so agents can inspect them through tools.

EXPLAINING A NON-LINEAR NUMBER

The drivers interact, so the agents explain with counterfactuals, not plugs

Why it doesn't add up

- PD is logistic: the same macro shock lifts a downgraded borrower's PD far more
- PD × LGD × EAD multiply, so credit and macro changes compound
- Weights act on stressed ECL: severe is $189M vs $95M baseline in August
- So any one-shot split depends on the order; a plugged residual hides it

Counterfactual walk: swap one input set, re-run the engine

July inputs

Exits

Balances & terms

New loans

Credit attributes

Macro, one variable at a time

Scenario weights

Model versions

Overlays

= August booked?

INTERACTION, MEASURED

	Walk	Shapley
Unemployment	$4.25M	$4.41M
CRE prices	$3.07M	$2.94M

Each step telescopes, so the walk sums exactly; the residual is real, vs an independent August run.

Reversed walk + exact Shapley (32 driver runs, 64 macro) show how much is interaction.

Agents report the driver with its robustness; the Reconciliation agent flags WARNING if methods disagree.

Results are kept per loan and per step, so portfolio and borrower drill-downs are exact sums of the same numbers.

WHO READS WHAT

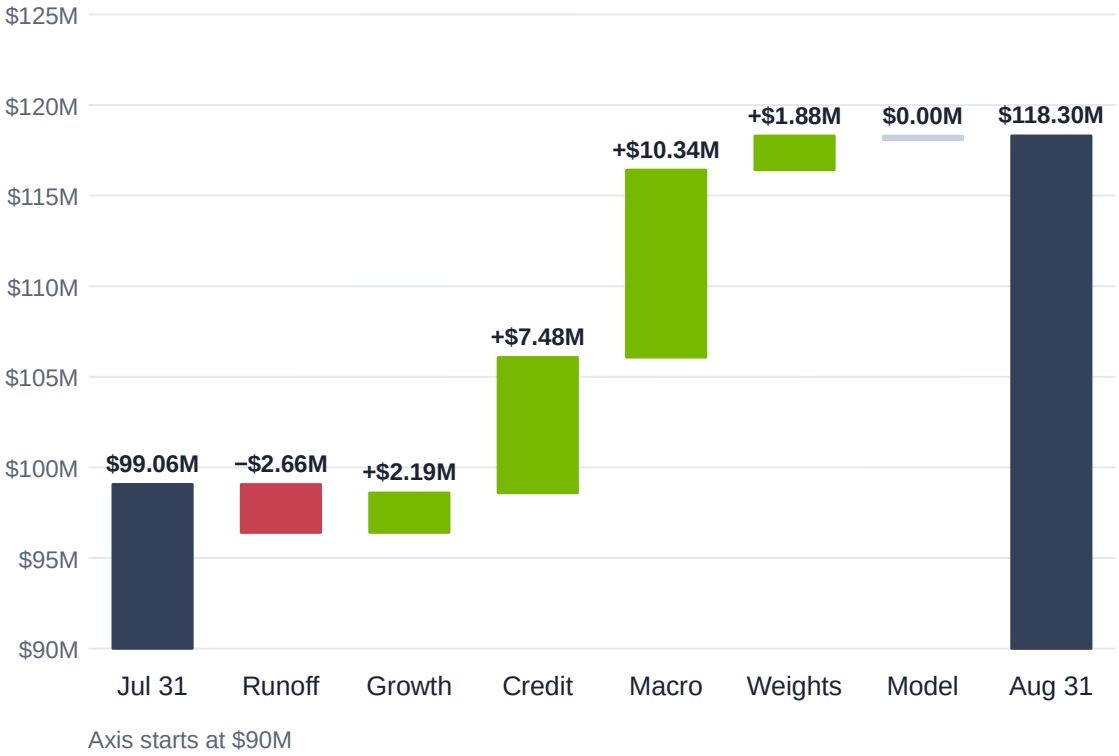
Each agent gets only the inputs and tools its job needs

Agent	Reads	Key tools	Tools
Orchestrator	the question; specialists' findings	ask_<specialist>, compare_allowance, attribution	—
Portfolio	balances, originations, exits, EAD	calculate_balance_change, calculate_new_originations, calculate_paydowns, calculate_ead_change	14
Credit	ratings, PD/LGD, DSCR, LTV, FICO, DPD	calculate_pd_migration, calculate_rating_migration, identify_top_contributors, compare_borrower_periods	15
Macro	scenario paths, weights, sensitivities	compare_macro_scenarios, calculate_variable_attribution, run_scenario_analysis	14
Model Risk	model inventory, parameters, overlays	compare_model_versions, compare_model_parameters, compare_management_overlays	12
Reconciliation	booked allowance vs engine re-runs	reconcile_allowance_change, calculate_portfolio_attribution	11
Quant	booked data, read-only sandbox	describe_datasets, run_python_analysis	5
Reporting	tool outputs and call log	build_attribution_workbook (Excel pack)	4
Regulatory	local guidance library, page-cited	search_regulation, get_regulatory_passage	4

Allow-lists in config/agents.yaml; a test keeps them in sync with the tools. All calls go through call_tool: validation, argument repair, provenance.

WHAT THE AGENTS FOUND

Macro and credit deterioration drove 93% of the increase



Axis starts at $90M

Reconciliation: PASS

Reported $19.24M · explained $19.24M · residual $0.00M (100.0%). Seven controls pass.

Robust to order

Macro stays the primary driver under the forward walk, the reversed walk and exact Shapley over all driver groups.

Model versions: $0.00M. The whole methodology bucket is the scenario-weight change.

DRILL-DOWN

From portfolio to borrower, matching the hidden ground truth

CRE

+$13.89M

72% of the change: macro +$5.69M, credit +$5.13M

Unemployment

+$4.41M

largest macro variable (Shapley share of the $10.34M macro effect); CRE prices lead inside CRE

CP_040408

1.8% → 8.1%

12-month PD of the top borrower: BB+ → B+, DSCR 1.33 → 1.16, LTV 0.68 → 0.74, watchlist

CRE_PD_v4

No change

same PD and LGD models both months; severe weight 10% → 12% added $1.88M

Ground truth check: primary driver macro ✓ portfolio CRE ✓ variable unemployment ✓ model changed: no ✓ largest borrower CP_040408 ✓

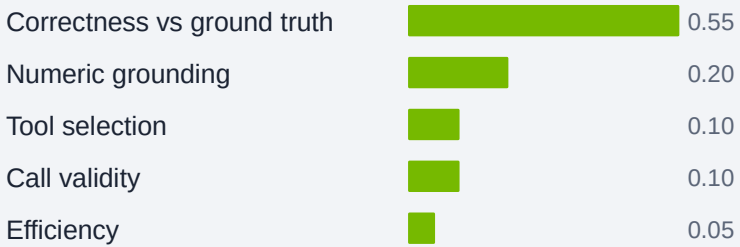
Ground truth is stored apart from the tools; a test checks that no tool or agent can read it.

HOW WE SCORE IT

15 validation questions, one verifier, same tools for every model

01 July allowance
02 August allowance
03 Dollar change
04 Percent change
05 Major drivers
06 Top portfolio
07 Top borrowers
08 Did PD rise?
09 Did LGD rise?
10 Did exposure grow?
11 Top macro variable
12 Scenario weights
13 Model change?
14 What happened in CRE?
15 Does it reconcile?

NeMo Gym verifier reward (0–1)



LOCAL NEMOTRON TEAM · 16 GB LAPTOP

0.892 **89% of frontier**

10 / 15 fully right · 0 failed runs · Nemotron 3 Nano 4B on Ollama

A give-up scores 0. Team score credits specialists' tool calls from the call log (first scored 0.832). Frontier: Claude Opus 5.5, identical prompt, tools, data, verifier.

RESULTS

A 3B-active Nemotron reaches 98% of frontier quality with the same tools



Lightning: 98% of frontier on one GPU. A 4B on a 16 GB laptop: 90%. Same 15 questions, tools and verifier; nothing leaves the machine.

Bar length = mean verifier reward (0–1). Held-out and LoRA rows use the property-shock story, which has different correct answers.

TOOL-CALLING RELIABILITY

Errors happen. The system recovers and shows its work

1.00
tool selection

0.974
numeric grounding

0.944
call validity

2 / 2
failed calls recovered on retry

Nemotron 3.5 Lightning in NeMo Gym, 15 questions, 23 tool calls.

Built into every call

- Dates and names normalised; JSON-string arguments repaired
- Typed errors with a hint the model uses to retry
- Every figure in the answer checked against tool outputs
- Provenance: call_id, parameters, timestamp, result hash

What the laptop 4B taught us

- Ollama's 4k default context silently cut the 11.8k-token prompt; with 32k and a warm-up the 4B scores 0.898 on all 15
- One give-up: tool calls written as text; now scored 0, not 0.988
- ~89 s per question on an M4, almost all model time: speed is the next gap

MULTI-AGENT TRACK

On a laptop 4B, the team ties the single agent and wins the hard questions

0.892
team, mean reward

0.898
single agent

10 / 15
fully right, both

192 s
vs 89 s per question

Team won the analytical questions

- Reconciliation: 1.00 vs 0.59
- Major drivers: 0.86 vs 0.74
- What happened in CRE: 0.82 vs 0.72
- Specialist prompts 3.5–5k tokens and 11–15 tools vs ~11.8k and 33

Team lost on a long hand-off

- Top borrowers: 0.35 vs 1.00 (853 s, the slowest delegation)
- Model change: 0.90 vs 1.00
- Each delegation is another local model call: about 2× the time on a laptop

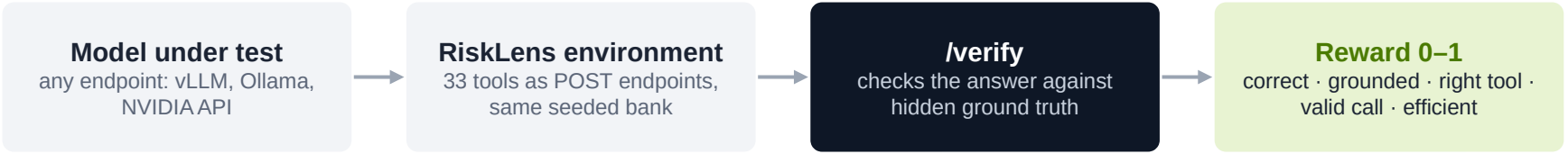
Next: route simple lookups to one specialist directly, keep the team for drivers, CRE and reconciliation, and run it on vLLM instead of a laptop.

Same model (Nemotron 3 Nano 4B, 32k, Ollama, M4 16 GB), same 15 questions and verifier. Team score credits specialists' tool calls from the call log.

NEMO GYM: REFEREE AND COACH

One Gym environment scores every model, then turns good runs into training data

REFEREE the same test for every model



Same verifier, all 15 questions: frontier 0.997 · Nemotron Lightning 0.977 (in Gym) · 4B laptop 0.898 single / 0.892 team (demo app)

COACH rewards decide what the model learns from



Story variants change the correct answers (credit-led, growth-led, weights shift, consumer stress); property shock is never trained on, so a higher score means the model learned the job, not the answers.

resources_servers/risklens_cecl (tools as endpoints + /verify); reward weights in config/gym.yaml. Every figure from the engine or the verifier.

FINE-TUNED & SPECIALIZED

LoRA makes fast, no-reasoning Nemotron call the tool first

Rollouts

540 Nemotron investigations in NeMo Gym across 5 story variants

Filter

Re-scored by the verifier; reward ≥ 0.95 kept: 401 rows

LoRA SFT

Rank 16, 2 epochs, 102 steps, 67 min on one A100 80 GB

Re-score

Held-out property-shock story, never trained on, 60 episodes

HELD-OUT STORY, REASONING OFF

0.751 → 0.939

Mean Gym reward, base → LoRA. Answers with no tool call: 14 / 60 → 0 / 60. Base made up figures (e.g. a $45.3M July allowance vs $97.49M in the engine).

Why it matters

- Near reasoning-mode quality (0.939 vs 0.964) with ~40% fewer output tokens (567 vs 970)
- Published: huggingface.co/gcav66/nemotron-3.5-lightning-risklens-lora
- Not yet: no gain with reasoning on (0.959 vs 0.964); CRE borrower drill-down not learned

WHY A BANK BUYS THIS

An explanation layer on top of the CECL engine the bank already owns

Close faster

The month-end “why” goes from days of cross-team work to a traced answer in minutes.

Audit-ready

Every figure ties to a tool call hash and reconciles to the booked allowance.

Data stays home

Small open models on the bank's own GPU or vLLM: no loan data sent to a frontier API.

No rip-and-replace

Reads the booked run from any vendor engine: CECL today, IFRS 9 in the same engine.

ILLUSTRATIVE VALUE, PER BANK

5 people × 3 days × 12 closes
≈ 1,440 hours a year
× $120 loaded hourly cost

≈ $173k / yr

of analyst time per bank, before audit findings avoided. Our assumptions, not survey data: replace with pilot measurements.

Buyers: allowance (ACL) owners in Finance and Credit Risk, Model Risk, internal audit. Sold alongside existing CECL vendors, not against them.

ROADMAP AND THE ASK

From hackathon to pilot

Now	Engine, 33 tools, agent team, NeMo Gym verifier, frontier reference, LoRA on Hugging Face, local 15-question benchmark
Next 30 days	Route lookups vs investigations; team on vLLM for speed; teach the CRE drill-down
Pilot	One bank's real extracts through Dataiku → RISKLENS_DATA_DIR; measure hours saved per close
Scale	IFRS 9 staging explanations, quarterly ACL committee pack, more story variants

We're asking for

- One design-partner bank with a CECL or IFRS 9 book
- GPU time to extend the fine-tune
- Feedback from allowance and model-risk practitioners

Small models don't need to know everything.

They need to know exactly which risk tool to call next.

RiskLens AI · Team Residual Zero
github.com/Luizkauffmann/Nvidia-Redhat-RiskLens

APPENDIX

Sources and provenance

1 Risk analytics market: $42.92B (2025), $47.26B (2026), $76.49B (2031), 10.11% CAGR. ResearchAndMarkets / Mordor Intelligence. link

2 70% of community banks use third-party vendors for CECL; 57% rely on third-party economic forecasts. ABA Banking Journal, Apr 27, 2026. link

3 77% of banks needed extra working days after year-end for IFRS 9 impairment (16% more than 10 days). Deloitte, Sixth Global IFRS Banking Survey, 2016, 91 banks. link

4 Fed Vice Chair for Supervision calls for CECL repeal at FASB roundtable. ABA Banking Journal, May 12, 2026. link

5 Project figures: RiskLens repo, demo_app/baselines/ (frontier reference, 15-question bank comparison) and evidence/ (Nemotron baseline, story variants, LoRA full run and held-out eval). All numbers from the deterministic engine or the Gym verifier. link

Illustrative value on slide 15 uses team assumptions, not survey data. Synthetic bank data: seed 20260831.